

Appendix D

III-V Device: Microwave IC*

Microwave IC devices are a specialized form of integrated circuit which are used as a high-frequency amplifier (2–18 GHz) for radar, telecommunications, telemetry, as well as octave and multi-octave amplifiers for use in electronic warfare systems. Microwave IC device manufacturers typically purchase the single-crystal GaAs substrates pre-doped from outside vendors (as do silicon device manufacturers). The major processing steps include liquid-phase epitaxial deposition, fabrication and non-fabrication processing similar to silicon device manufacturing. Processing steps which warrant additional description from the LED process are discussed.

Microwave IC production is comprised of three separate steps (see Table D1) in addition to surface preparation:

1. Fabrication of the GaAs FET (field-effect transistor) using liquid-phase epitaxy (LPE)
2. Hybrid circuit fabrication on a ceramic substrate
3. Final module assembly of the GaAs FET on the hybrid ceramic substrate

* This information is compiled primarily from references F1, F2, and F3 shown in Appendix F.

Table D1. III-V Device—Microwave IC

GaAa TRANSISTOR FABRICATION

I. SUBSTRATE PREP

- a. Lapping
- b. Polishing

II. EPITAXIAL

- a. Pre-clean
- b. LPE Growth

III. DEVICE FABRICATION

- a. Photolithographic
- b. Wet Etching
- c. Plasma Ashing
- d. Metallization
- e. Backlapping
- f. Scribe and Break

HYBRID CIRCUIT FABRICATION

IV. CERAMIC SUBSTRATE

- a. Clean and Dip
- b. Plasma Ashing
- c. Metallization
- d. Photolithographic
- e. Wet Etching

MODULE ASSEMBLY

V. ATTACHMENT

- a. Brazing
 - b. Bonding
 - c. Forming
-

GaAs TRANSISTOR FABRICATION

Substrate Preparation

The chromium doped GaAs substrate, supplied by an outside vendor, must be prepared prior to LPE.

Lapping. Wafers are wax-mounted and lapped using a wet slurry mixture of silicon carbide abrasive, ethylene glycol, and water.

Polishing. Wafers are polished using a solution of 5% chlorine and water, with demounting performed on a hot plate.

Epitaxial

Pre-Clean. Prior to LPE deposition of a GaAs layer, the following sequence of cleaning steps are performed:

1. Detergent and water degreasing
2. Acetone, isopropyl alcohol, and de-ionized water rinse
3. Sulfuric acid, hydrogen peroxide, and water etch (5:1:1)
4. Fifty percent hydrochloric acid and water etch
5. De-ionized water rinse and nitrogen dry

LPE Growth. In liquid-phase epitaxial growth, a layer of n-type GaAs (doped with tin) is grown on the exterior surface of the wafer. The system consists of:

1. A moveable cylindrical furnace capable of maintaining 720°C temperature
2. A quartz reactor chamber which can be pressured to 0.3 psi
3. A graphite holder which contains separate chambers for the doped and undoped GaAs melts and GaAs wafers to be deposited
4. A pullrod to physically move the wafers from one melt to the next
5. A chamber connector plate which allows pressurization and routes hydrogen/nitrogen gas flow

Pre-weighed quantities of gallium, arsenic, and tin dopant are added to one chamber in the graphite holder, with small chunks of undoped GaAs added to the other end, and the wafers on a lower chamber, in the enclosed quartz tube. The quartz tube is connected to the metal connector plate and the tube guided part way into the cylindrical furnace—far enough to heat the multi-compartment boat. The metal GaAs melt is allowed to equilibrate and then the melt is positioned over the substrate and the furnace temperature is lowered to form the n-type GaAs layer.

Device Fabrication

Photolithographic. The standard photoresist, aligning/exposing, developing, and stripping process is utilized as in silicon device processing.

Wet Etching. Various mixtures of wet chemical acid solutions are used, such as:

1. EPI-Mesa etch: 10% hydrofluoric acid, 10% hydrogen peroxide, and de-ionized water
2. Aluminum etch: mixture of 50% sodium and zinc cyanide with organic sulfonated benzene and de-ionized water
3. Gold lift-off: acetone and isopropyl alcohol in an ultrasonic bath

Plasma Ashing. A barrel-type plasma asher utilizing oxygen is used for descumming and ashing. A radio-frequency source at 13.56 MHz was used.

Metallization. A gold deposition is performed using an e-beam evaporator and layers of titanium, platinum, and gold are sequentially evaporated. An aluminum evaporation is also performed using an e-beam unit.

A glassivation layer of silicon dioxide is evaporated using a combination filament/e-beam evaporator, with silicon monoxide and oxygen providing the chemical species.

Backlapping. A backlap is performed utilizing the same method as was described previously. A detergent rinse is used as a final cleaning step.

Scribe and Break. The individual FETs on the GaAs wafer are diamond-scribed and broken on a special mounting apparatus. The individual FET die will then go to module assembly for final mounting on the ceramic hybrid circuit.

HYBRID CIRCUIT FABRICATION

Ceramic Substrate

A ceramic (alumina, Al_2O_3) substrate is used for fabricating and attaching the various components necessary to produce a microwave IC

device. This includes metallization of the substrate for attachment of the resistors, capacitors and the GaAs transistor in module assembly.

Clean and Dip. The following sequence of steps is used:

1. Methyl ethyl ketone (MEK), organic stripper (phenol, sulfonic acid, and chlorobenzenes) and de-ionized water rinse
2. Ammonium hydroxide pre-conditioning dip
3. Isopropyl alcohol (IPA), trichloroethylene (TCE), acetone dip, and IPA vapor degreasing

Plasma Ash. A barrel-type plasma asher is used to clean the unmetallized ceramic substrate.

Metallization. An RF sputtering system is used to vacuum deposit layers of tantalum and gold on the substrate.

Wet Etching. Prior to etching the deposited layers of tantalum and gold, a photolithographic sequence is used to define the metal circuit pattern desired. The etch solutions are:

1. Gold: potassium iodide at 50°C
2. Tantalum: mixture of hydrofluoric and nitric acids and de-ionized water.

The photoresist is then stripped off, using an organic stripper, as previously described.

MODULE ASSEMBLY

Attachment

The completed GaAs FET and metallized ceramic substrate, in addition to the resistors and capacitors, undergo a final attachment step.

Brazing. A pre-form type brazing material made of 88% gold and 12% germanium is used to non-flux braze various components to the ceramic substrate.

Bonding. The GaAs FET transistor is epoxy-bonded to the substrate using a one-component epoxy resin. Then the gold lead wires, which electrically interconnect the FET to the substrate and components, are compression-bonded at 250°C.

Forming. The completed device then has a metal container housing lid which is sealing soldered on, using 60% tin, 40% lead solder. This is done in a forming gas furnace containing $\leq 15\%$ hydrogen and the balance nitrogen. The furnace provides the dual function of soldering and annealing the completed device. Various quality assurance (QA) tests are performed, and a final tuning of the frequency of the amplifier is done prior to shipment.